QA Test Cases: Geometry Programs

1. Triangle Program (`check_triangle`)

Test Case ID	Title / Description	Preconditions / Postconditions	Test Steps	Test Data	Expected Results	Actual Results	Pass / Fail Criteria
TC-TRI-01	**Valid Equilateral Triangle**	**Pre:** Program is running, user selects Triangle check. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	5, 5, 5	Prints "It is an equilateral triangle."	(Pending execution)	Pass if the exact expected string is printed, Fail otherwise.
TC-TRI-02	**Valid Isosceles Triangle**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	5, 5, 3	Prints "It is an isosceles triangle."	(Pending execution)	Pass if the exact expected string is printed.
TC-TRI-03	**Valid Scalene Triangle**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	3, 4, 5	Prints "It is a scalene triangle."	(Pending execution)	Pass if the exact expected string is printed.
TC-TRI-04	**Invalid Triangle (Sum equals third)**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	2, 3, 5	Prints "Error: The given sides cannot form a valid triangle."	(Pending execution)	Pass if the error message for invalid triangle is printed.

Test Case ID	Title / Description	Preconditions / Postconditions	Test Steps	Test Data	Expected Results	Actual Results	Pass / Fail Criteria
TC-TRI-05	**Invalid Triangle (Sum less than third)**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	1, 2, 10	Prints "Error: The given sides cannot form a valid triangle."	(Pending execution)	Pass if the error message for invalid triangle is printed.
TC-TRI-06	**Zero Length Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	0, 4, 5	Prints "Invalid input: Please make sure all numbers are greater than zero."	(Pending execution)	Pass if the script rejects the zero input correctly.
TC-TRI-07	**Negative Length Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	4, -2, 5	Prints "Invalid input: Please make sure all numbers are greater than zero."	(Pending execution)	Pass if the script rejects negative values.
TC-TRI-08	**All Zeroes Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side 2. Enter second side 3. Enter third side	0, 0, 0	Prints "Invalid input: Please make sure all numbers are greater than zero."	(Pending execution)	Pass if the script rejects all zeroes.
TC-TRI-09	**Non-integer Input (Characters)**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side	"A"	Prints "Error: Inputs must be integers." and exits function.	(Pending execution)	Pass if the exception is caught and error is printed.

Test Case ID	Title / Description	Preconditions / Postconditions	Test Steps	Test Data	Expected Results	Actual Results	Pass / Fail Criteria
TC-TRI-10	**Float Input (Decimals)**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter first side	"3.5"	Prints "Error: Inputs must be integers."	(Pending execution)	Pass if it rejects the decimal string format since `int()` throws an error on floats in string form.

2. Quadrilateral Program (`check_quadrilateral`)

Test Case ID	Title / Description	Preconditions / Postconditions	Test Steps	Test Data	Expected Results	Actual Results	Pass / Fail Criteria
TC-QUAD-01	**Valid Square**	**Pre:** Program is running, user selects Quadrilateral check. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	4, 4	Prints "It is a square."	(Pending execution)	Pass if the square identification message is printed.
TC-QUAD-02	**Valid Rectangle**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	4, 6	Prints "It is a rectangle."	(Pending execution)	Pass if the rectangle identification message is printed.
TC-QUAD-03	**Zero Length Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	0, 5	Prints "Invalid input: Please make sure both length and width are greater than zero."	(Pending execution)	Pass if it properly catches the zero dimension.
TC-QUAD-04	**Zero Width Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	5, 0	Prints "Invalid input: Please make sure both length and width are greater than zero."	(Pending execution)	Pass if it properly catches the zero dimension.
TC-QUAD-05	**Negative Length Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2.	-3, 6	Prints "Invalid input: Please make sure	(Pending execution)	Pass if negative dimensions are rejected.

Test Case ID	Title / Description	Preconditions / Postconditions	Test Steps	Test Data	Expected Results	Actual Results	Pass / Fail Criteria
			Enter width		both length and width are greater than zero."		
TC-QUAD-06	**Negative Width Input**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	6, -4	Prints "Invalid input: Please make sure both length and width are greater than zero."	(Pending execution)	Pass if negative dimensions are rejected.
TC-QUAD-07	**Both Negative Dimensions**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	-5, -5	Prints "Invalid input: Please make sure both length and width are greater than zero."	(Pending execution)	Pass if both negatives are successfully caught.
TC-QUAD-08	**Both Zero Dimensions**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length 2. Enter width	0, 0	Prints "Invalid input: Please make sure both length and width are greater than zero."	(Pending execution)	Pass if both zero bounds are successfully caught.
TC-QUAD-09	**Non-integer Input (Characters)**	**Pre:** Program is running. **Post:** Program	1. Enter length	"Square"	Prints "Error: Inputs	(Pending execution)	Pass if the invalid string input is caught by

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		returns to main menu.			must be integers."		the try/except block.
TC-QUAD-10	**Float Input (Decimals)**	**Pre:** Program is running. **Post:** Program returns to main menu.	1. Enter length	"5.5"	Prints "Error: Inputs must be integers."	(Pending execution)	Pass if decimal numbers are caught by the type cast attempt.